

MMC 개발 관련 자료

07.03.28

반도체부품개발센터 G1사업개발그룹

1. 기능

1. MMC spec. v3.3, v4.0/v4.1 지원
2. MMC command : 0, 2, 4, 5, 6, 7, 8 지원
3. Firmware ISP(In System Programming) function, firmware upgrade 지원
4. High performance, low power
 - Single channel performance up to 20 MB/s
 - Dual channel performance up to 42 MB/s
 - Whole card standby current < 150 μ A
 - Whole card operation current < 30 mA
5. Flash interface
 - Single channel 8 bit/16 bit flash interface
 - Dual channel 8 bit flash interface
 - Each channel support up to 4 flash chips
 - Total memory capacity up to 4 GBytes

2. MMC Host 연동

1. 768개의 제품과 호환성 검사를 해서 이상이 없는 것으로 보아 MMC Spec.을 따라 제조한 것으로 생각되며 호환성 문제가 없을 것으로 사료됨
2. 52 MHz 로 동작하는 MMC 인터페이스에 timing margin을 확보하지 못한다는 것은 문제가 있음
3. 향후 다양한 제품을 통해 검증은 필요함

	Memory Card Type	MMC Card
1	USB Card Reader	171
2	PDA	123
3	DSC/DVC	281
4	SD/MMC to PCMCIA Adapter	6
5	Notebook PC & PC	48
6	SD/MMC Mobile Phone	14
7	Audio Player	45
8	Printer	36
9	Others	44
Total		768

3. Flash 연동 (1)

Vendor	SLC/MLC	CapacitV	Part Number	SM265
			HW Version	AB
			ISP Version	F0922
Hynix	SLC	256Mbit	HY27US08561M	Y
Hynix	SLC	512Mbit	HY27US08121M	Y
Hynix	SLC	1Gbit	HY27US08121M	Y
Hynix	SLC	256bit (1.8V)	HY27SS08561M	
Hynix	SLC	2Gbit	HY27UB082G4M (2K/page)	Y
Hynix	SLC	1Gbit	HY27UF081G2M (2K/page)	Y
Hynix	SLC	1Gbit (70nm, mono die)	HY27UF081G2A (SLC)	Y
Hynix	SLC	2Gbit	HY27UG082G2M (2K/page)	Y
Hynix	SLC	4Gbit	HY27UH084G2M (2K/page)	Y
Hynix	SLC	2Gbit	HY27UF082G2M	Y
Hynix	SLC	4Gbit	HY27UG084G2M	Y
Hynix	SLC	8Gbit	HY27UH088G2M	Y
Hynix	SLC	4Gbit	HY27UF084G2M	Y
Hynix	SLC	8Gbit	HY27UG088G5M	Y
Hynix	SLC	18Gbit	HY27UH08AG5M	Y
Hynix	MLC	4Gbit (Single Die)	HY27UT084G2M (Big Block MLC) *Note	Y
Hynix	MLC	8Gbit (DDP)	HY27UU088G5M (Big Block MLC) *Note	Y
Hynix	MLC	18Gbit (QDP)	HY27UV08AG5M (Big Block MLC) *Note	Y
Hynix	MLC	32Gbit	HY27UW08BGFM (Big Block MLC) *Note	
Hynix	SLC	1Gbit (70nm)	HY27UF081G2A (SLC)	Y
Hynix	SLC	2Gbit (70nm)	HY27UF082G2A (SLC)	Y
Infenion	SLC	512Mbit	HYF33DS512800ATC	
Infenion	SLC	512Mbit	HYF31DS512805BTC	Y
Infenion	SLC	1Gbit	HYF33DS1G80OCT1	F1003
Intel	SLC	2Gbit (90nm)	MT29F02G08AANA2	Y
Intel	SLC	2Gbit (72nm)	29F02G08AANB3	Y
Intel	SLC	4Gbit (72nm) DDP	29F04G08BANB3	Y
Intel	SLC	8Gbit QDP (72nm)	29F08G08FANB3	Y
Intel	SLC	8Gbit (72nm) DDP	29F08G08BANB1	Y
Intel	SLC	4Gbit (72nm) Single Die	29F04G08AANB1	Y
Intel	SLC	8Gbit (72nm) DDP	29F08G08CANB1	Y
Intel	SLC	18Gbit (72nm) QDP	29F18G08FANB1	Y
Intel	MLC	8Gbit (72nm) mono	29F08G08AAMB1	Y
Intel	MLC	18Gbit (72nm) DDP	29F18G08CAMB1	Y
Micron	SLC	2Gbit	MT29F2G08AAB	Y
Micron	SLC	4Gbit	MT29F4G08BAB	Y
Micron	SLC	8Gbit	MT29F8G08FAB	Y
Micron	SLC	4Gbit	MT29F4G18BAB (18bit)	Y
Micron	SLC	1Gbit (1.8V)	MT29F1G08ABAWP (ES sample)	
Micron	SLC	2Gbit (1.8V, 70nm)	MT29F2G08ABC (ES sample)	

3. Flash 연동 (2)



Vendor	SLC/MLC	CapacitV	Part Number	SM265
			HW Version	AB
			ISP Version	F0922
Micron	SLC	2Gbit (1.8V, 72nm)	MT29F2G08ABC (MP sample)	
Micron	SLC	2Gbit (3.3V, 72nm)	MT29F2G08AAC (MP sample)	Y
Micron	SLC	2Gbit (16 bit)	MT29F2G16AAB	Y
Micron	SLC	4Gbit (16 bit)	MT29F4G16AAB	Y
Micron	SLC	2Gbit (1.8V, 72nm)	MT29F2G08A5C	
Micron	SLC	2Gbit (3.3V, 72nm)	MT29F2G08AAC (SLC)	Y
Micron	SLC	4Gbit (72nm) Single Die	MT29F4G08AAA (SLC)	Y
Micron	SLC	8Gbit (72nm) DDP	MT29F8G08BAA (SLC)	Y
Micron	SLC	16Gbit (72nm) QDP	29F16G08FAA (SLC)	Y
Micron	MLC	4Gbit (72nm) Single Die	MT29F4G08MAAWP-R1 ES	Y
Micron	MLC	8Gbit (72nm) DDP	MT29F8G08MAAWP-R1 ES	Y
Renesas	MLC	1Gbit	HN29V1G91T-30V(MLC)	Y
Renesas	MLC	1Gbit	HN29V1G91T-30(MLC)	Y
Renesas	MLC	2Gbit	HN29V2G74W30	Y
Renesas	MLC	4Gbit	R1FV04G13R(14R)	
Power Chip	MLC	4Gbit	P1U4GR30CT-G45CA	
Samsung	SLC	1Gbit	K9T1G08U0M	Y
SamSung	SLC	512Mbit	K9K1208U0C	Y
SamSung	SLC	512Mbit (1.7~1.95V)	K9K1208Q0C	
SamSung	SLC	512Mbit	K9F1208U0B	Y
SamSung	SLC	512Mbit (1.7~1.95V)	K9F1208Q0B	
SamSung	SLC	512Mbit	K9F1208U0A	Y
SamSung	SLC	512Mbit	K9F1208U0M	Y
SamSung	SLC	256Mbit	K9F5608U0C	Y
SamSung	SLC	256Mbit (1.7~1.95V)	K9F5608Q0C	
SamSung	SLC	256Mbit	K9F5608U0B	Y
SamSung	SLC	256Mbit (1.7~1.95V)	K9F5608Q0B	
SamSung	SLC	256Mbit	K9F5608U0A	Y
SamSung	SLC	128Mbit	K9F2808U0C	Y
SamSung	SLC	128Mbit	K9F2808U0B	Y
SamSung	SLC	128Mbit (1.7~1.95V)	K9F2808Q0B	
SamSung	SLC	64Mbit	K9F8408U0C	Y
SamSung	SLC	64Mbit (1.7~1.95V)	K9F8408Q0C	
SamSung	SLC	64Mbit	K9F8408U0B	Y
SamSung	SLC	1Gbit(1.65~1.95V)	K9F1G08R0A-CCB0	
SamSung	SLC	512Mbit(1.7~1.95V)	K9F1208Q0C	
SamSung	SLC	256Mbit(1.7~1.95V)	K9F5608Q0C	
SamSung	SLC	8Gbit	K9W8G08U1M	Y
SamSung	SLC	2Gbit	K9K2G08U0M	Y
SamSung	SLC	4Gbit	K9K4G08U0M	Y
SamSung	SLC	2Gbit	K9F2G08U0M	Y
SamSung	SLC	1Gbit	K9F1G08U0A	Y

3. Flash 연동 (3)



Vendor	SLC/MLC	CapacitV	Part Number	SM265
			HW Version	AB
			ISP Version	F0922
SamSung	SLC	1Gbit	K9K1G08U0A	Y
SamSung	SLC	1Gbit (1.7~1.95V)	K9F1G08Q0A	
SamSung	SLC	1Gbit	K9K1G08U0M	Y
SamSung	SLC	1Gbit	K9F1G08U0M	Y
SamSung	SLC	1Gbit (1.7~1.95V)	K9F1G08Q0M	
SamSung	SLC	2Gbit	K9E2G08U1M	Y
SamSung	SLC	4Gbit	K9F4G08U0M	Y
SamSung	SLC	8Gbit	K9K8G08U0M	Y
SamSung	SLC	4Gbit	K9F4G08U0A	Y
SamSung	SLC	8Gbit	K9K8G08U0A	Y
SamSung	SLC	16Gbit	K9WAG08U1M	Y
SamSung	SLC	16Gbit	K9WAG08U1A	Y
SamSung	SLC	32Gbit	K9NBG08U5M	
SamSung	SLC	2Gbit (63nm) Single Die	K9F2G08U0A (2 plane)	Y
SamSung	MLC	4Gbit	K9G4G08U0M (MLC)	Y
SamSung	MLC	8Gbit	K9L8G08U0M (MLC)	Y
SamSung	MLC	16Gbit	K9HAG08U1M (MLC)	Y
SamSung	MLC	32Gbit	K9MBG08U5M (MLC)	
SamSung	MLC	8Gbit (63nm) Single Die	K9G8G08U0M (MLC)	Y
SamSung	MLC	16Gbit (63nm) DDP	K9LAG08U0M (MLC)	Y
SamSung	MLC	32Gbit (63nm) QDP	K9HBG08U1M (MLC)	Y
SamSung	MLC	4Gbit (63nm) Single Die	K9G4G08U0A (MLC)	Y
SanDisk	SLC	1Gbit	SDTNGEHE0-1024 (Big Block)	Y
SanDisk	MLC	1Gbit	SDTNFCH - 1024 (MLC)	Y
SanDisk	MLC	1Gbit	SDTNGCHE0-1024 (MLC)	Y
SanDisk	MLC	2Gbit	SDTNGCHE0-2048 (MLC)	Y
SanDisk	MLC	4Gbit	SDTNGCHE0-4096 (MLC)	Testing
STM	SLC	256Mbit	NAND256W3A2AN6	Y
STM	SLC	512Mbit	NAND512W3A3AN1	Y
STM	SLC	512Mbit	NAND512W3A2BN6	Y
STM	SLC	512Mbit	NAND512W3A0AN6 (No interleaving)	Y
STM	SLC	512Mbit	NAND512W3B2AN6E	Y
STM	SLC	512Mbit	NAND512W3B2AN6 (Force ID)	Y
STM	SLC	1Gbit	NAND1GW3A0AN6 (No interleaving)	Y
STM	SLC	1Gbit	NAND01GW3A3AN6	Y
STM	SLC	1Gbit	NAND01GW3A0AN6	Y
STM	SLC	1Gbit	NAND01GW3B2AN6 (2K/page)	Y
STM	SLC	2Gbit	NAND02GW3B2AN6 (2K/page)	Y
STM	SLC	2Gbit	NAND02GW3B2BN6, D/C: 522	Y
STM	SLC	4Gbit	NAND04GW3B2AN6 (SLC)	Y
STM	SLC	4Gbit (70nm, mono die)	NAND04GW3B2B (SLC)	Y
STM	SLC	8Gbit (70nm, DDP)	NAND08GW3B2A (SLC)	Y
STM	SLC	1Gbit	NAND01GW3B2BN6	Y

3. Flash 연동 (4)



Vendor	SLC/MLC	CapacitV	Part Number	SM265
			HW Version	AB
			ISP Version	F0922
STM	MLC	4Gbit (90nm) Single Die	NAND04GW3C2AN1(MLC)	Y
Toshiba	SLC	128Mbit	TC58DVM72A1FT	Y
Toshiba	SLC	128Mbit	TC58DVM72F1FT	Y
Toshiba	SLC	256Mbit	TC58DVM82A1FT	Y
Toshiba	SLC	256Mbit	TC58DVM82A1XBJ1	Y
Toshiba	SLC	256Mbit	TC58DVM82F1FT	Y
Toshiba	SLC	512Mbit	TC58DVM92A1FT	Y
Toshiba		64Mbit	TC58V64BFT	Y
Toshiba		1Gbit	TC58DVG02A1FT	Y
Toshiba		1Gbit	TC58NVG0S3AFT	Y
Toshiba		2Gbit	TH58NVG1S3AFT	Y
Toshiba		1Gbit	TC58NVG0D4BTG00	Y
Toshiba		2Gbit	TC58NVG1D4BTG00	Y
Toshiba	MLC	4Gbit (90nm)	TC58NVG2D4BTG00(MLC)	Y
Toshiba	MLC	4Gbit (90nm)	TC58NVG2D4BFT00(MLC)	Y
Toshiba	MLC	8Gbit (90nm)	TH58NVG3D4BTG00(MLC)	Y
Toshiba	MLC	8Gbit (70nm) Single die	TC58NVG3D4CFT00(MLC)	Y
Toshiba	MLC	8Gbit (70nm)	TC58NVG3D4CTG00(MLC)	Y
Toshiba	MLC	16Gbit (70nm)	TH58NVG4D4CTG00(MLC)	Y
Toshiba	MLC	2Gbit	TC58DVG14B1FT00(MLC)	Y
Toshiba	MLC	1Gbit	TC58DVG04B1FT00(MLC)	Y
Toshiba	MLC	32Gbit (70nm)	TH58NVG5D4CTG20	Testing

4. 검토 대상 Flash



Vender	SLC/MLC	CapacitV	Part Number
Hynix	SLC	4Gbit	HY27UH084G2M(2K/page)
Hynix	SLC	4Gbit	HY27UG084G2M
Hynix	SLC	8Gbit	HY27UH088G2M
Hynix	SLC	4Gbit	HY27UF084G2M
Hynix	SLC	8Gbit	HY27UG088G5M
Hynix	SLC	16Gbit	HY27UH08AG5M
Hynix	MLC	4Gbit(Single Die)	HY27UT084G2M (Big Block MLC)
Hynix	MLC	8Gbit(DDP)	HY27UU088G5M (Big Block MLC)
Hynix	MLC	16Gbit(QDP)	HY27UV08AG5M (Big Block MLC)
Hynix	MLC	32Gbit	HY27UW088GFM (Big Block MLC)
Samsung	SLC	8Gbit	K9W8G08U1M
Samsung	SLC	4Gbit	K9K4G08U0M
Samsung	SLC	4Gbit	K9F4G08U0M
Samsung	SLC	8Gbit	K9K8G08U0M
Samsung	SLC	4Gbit	K9F4G08U0A
Samsung	SLC	8Gbit	K9K8G08U0A
Samsung	SLC	16Gbit	K9WAG08U1M
Samsung	SLC	16Gbit	K9WAG08U1A
Samsung	SLC	32Gbit	K9NBG08U5M
Samsung	MLC	4Gbit	K9G4G08U0M (MLC)
Samsung	MLC	8Gbit	K9L8G08U0M (MLC)
Samsung	MLC	16Gbit	K9HAG08U1M (MLC)
Samsung	MLC	32Gbit	K9MBG08U5M (MLC)
Samsung	MLC	8Gbit (63nm) Single Die	K9G8G08U0M (MLC)
Samsung	MLC	16Gbit (63nm) DDP	K9LAG08U0M (MLC)
Samsung	MLC	32Gbit (63nm) QDP	K9HBG08U1M (MLC)
Samsung	MLC	4Gbit (63nm) Single Die	K9G4G08U0A (MLC)
STM	SLC	4Gbit	NAND04GW3B2AN6 (SLC)
STM	SLC	4Gbit (70nm, mono die)	NAND04GW3B2B (SLC)
STM	SLC	8Gbit (70nm, DDP)	NAND08GW3B2A (SLC)
STM	MLC	4Gbit (90nm) Single Die	NAND04GW3C2AN1 (MLC)

5. Flash 동작 속도

1. 동작 속도

- tWC, tRC : 37.5 ns (P.25)
- 26 MB 이상이 나올 수 없는 구조

2. 검토 사항

- 해당 Flash memory 마다 tWC, tRC 를 바꿀 것인지 검토
- 주력 제품에서 가장 속도가 낮은 tWC, tRC 로 고정하면 다양한 Flash 지원 못함

3. K9K8G08U0M 측정 속도

Mode	Sequential Write (MB/s)	Sequential Read (MB/s)
Single	10.75	18.95
Twin	20.73	36.84
Single-Interleave	13.66	20.38
Twin-Interleave	27.32	40.55

6. Flash 검토 항목

1. SLC/MLC : 확인 !!
 - 서로 다른 동작 특성 여부 확인
 - 전원 ON 시 셋팅
2. AG-AND
 - 동작 특성 확인
3. Interleave 방식 지원

7. 단계별 산출물

1. Pin description 자료
2. System 클럭도 & 설명서
3. SoC 전체 블록도 & 설명서
4. 세부 블록 설계도 & 설명도
5. SoC chip source code
6. Simulation testbench file
7. Synthesis timing constraint file
8. DC synthesis & Simulation script file
9. 관련 자료

8. Test 방안

